Springboard
Guided Capstone
Big Mountain Resort

Springboard

Board & Management Challenges

Because the Management of Big Mountain Resort (BMR) foresees an opportunity to challenge the status quo of its financial performance, we have considered the underlying data within which the resort and its peer group operate. From a population of 330 ski resorts in the United States, BMR faces both opportunities and challenges as it considers price changes for its 2026 – 2027 ski season.

The challenge is to identify trends, metrics, and predictive modeling outcomes that support the Board and Management's decision to raise ticket prices. Our findings support the competitive advantages and unique attributes that BMR maintains over its competitors and options for pricing, venue updates, and operational enhancements; thereby, advancing the resort's objective of maximizing pricing while mitigating the risk of losing season-pass and day-pass customers.

Board & Management Challenges

We employed the following data science techniques in pursuit of determining the comparative strengths of BMR and those factors that serve as the defining attributes that drive ski lift ticket prices:

- Principal Component Analysis (PCA)
- Peer Segment Analysis Using Mean, Z-Score
- Correlation Analysis
- Cross Validation Score
- Random Forest Regression
- Scenario Analysis
- Analysis of Data Deficiency

We note that these methods of analysis had uneven outcomes and predictive value. To that end, we have chosen only those methods that are most likely to prove valuable in your consideration of the relative strengths of the resort against peers.

Recommendations & Key Findings

Executive Summary

Our analysis and modeling suggest that Big Mountain Resort has flexibility to increase its lift ticket pricing in the 2026 – 2027 ski season. The basis of our recommendation stems from the following methods of data science extraction:

- Data Set Analysis – BMR's attributes were compared with its peer-group using statistical measures in a segment framework. BMR outperformed competitors.
- Data Set Strengths Compared with Correlation Analysis – We note variables that that support adult weekend ticket prices. BMR had high correlation to all five.
- Modeling Results – Finally, our model suggests the expected adult weekend lift-ticket price for the resort is $95.87, compared with its current price of $81.00.
- Cross Validation (CV) – Utilizing CV testing, or how well the model performs on unseen data, additional variables are unlikely to support differing conclusions.

Modeling Results & Analysis: Data Set Analysis

Data Set Analysis – BMR's attributes compared with its peers, which prove favorable.

Comparison of Big Mountain Resort Versus All Resorts in Sample Population

Segment	Database Field	BMR Actual	All Resorts Mean	BMR - All Resort Mean	All Resorts Std. Dev.	Z-Score*
Elevation & Terrain Scale	Summit Elevation	6,817	4,585	2,232	3,739	0.60
	Vertical Drop	2,353	1,212	1,141	947	**1.20**
	Base Elevation	4,464	3,371	1,093	3,121	0.35
	Total Runs	105	48	57	46	**1.24**
	Terrain Parks	4.0	2.8	1.2	2.0	0.60
	Longest Run (mile)	3.3	1.43	1.87	1.15	**1.63**
	Skiable Terrain (acres)	3,000	656	2,344	1,094	**2.14**
Lift Infrastructure	Trams	0	0.17	–0.17	0.56	**–0.30**
	Fast Eights	0	0.01	–0.01	0.08	**–0.13**
	Fast Sixes	0	0.19	–0.19	0.65	**–0.29**
	Fast Quad	3	1.01	1.99	2.2	0.90
	Quad	2	0.93	1.07	1.31	0.82
	Tripple	6	1.49	4.51	1.6	**2.82**
	Double	0	1.84	–1.84	1.81	**–1.02**
	Surface	3	2.62	0.38	2.06	0.18
	Total Chairs	14	8.25	5.75	5.8	0.99
Snow Conditions & Related Capacity	**Snow Making (acres)**	600	173	427	261	**1.64**
	Average Snowfall	333	185	148	136	1.09
	Night Skiing (acres)	600	98	502	99	**5.07**
Operating Season & Resort History	Days Open Last Year	123	115	8	35	**0.23**
	Projected Days Open	123	120	3	31	**0.10**
	Year Open	72	64	8	110	0.07
Ticket Pricing	Adult Weekday Ticket	$81	$58	$23	$26	**0.88**
	Adult Weekend Ticket	$81	$64	$17	$25	**0.68**
Resort Identification & Geo Info	Name of Resort	BMR	N/A	N/A	N/A	N/A
	Region	Montana	N/A	N/A	N/A	N/A
	State	Montana	N/A	N/A	N/A	N/A

*Z-Score – Calculated per ((BMR Actual – (minus) All Resorts Mean) / All Resorts Std. Dev.), with:
0.00 – Indicating equal to all resorts mean / +1.00 – Indicating one Stan. Dev. above mean / -1.00 – Indicating one Stan. Dev. below mean

Modeling Results & Analysis: Data Set Vs. Correlation Analysis

Data Set Strengths Compared to Correlation Analysis – From our correlation analysis, we note five variables that appear most positively correlated with adult weekend ticket prices, and BMR's relative strength in these categories, based on z-scores:

Rank of Five Highest Variables/Attributes to Adult Weekend Ticket Prices & BMR's Z-Score				
Rank	**Variable/Attribute**	**Correlation**	**BMR's Z-Score**	**Comparative Value**
1	Runs	0.7569	1.24	> 1 Stan. Dev. Above Peers
2	Fast Quads	0.7314	0.9	~ 1 Stan. Dev. Above Peers
3	Vertical Drop	0.7133	1.2	> 1 Stan. Dev. Above Peers
4	Snow Making (acres)	0.6958	1.64	> 1 Stan. Dev. Above Peers
5	Total Chair Lifts	0.6544	0.99	~ 1 Stan. Dev. Above Peers

The data indicates that those attributes that have the highest correlation to adult weekend ticket prices are categories that BMR enjoys comparative strength when compared to the peer-group. This provides BMR with pricing optionality, given other changes to operations and venue.

Modeling Results & Analysis: Modeling Results

Modeling Results – The model helps reinforce our analysis and recommendations to the management team by identifying resort attributes most strongly associated with ticket-price power. Based on the train/test split, cross-validation scoring, and random forest regressor feature-importance results, the strongest predictive factors are:

- Number of fast quad lifts
- Total ski runs
- Snow-making acreage/coverage
- Mountain’s vertical drop

We caution that the model’s predictive value does not necessarily prove causation, and, thus, conclusions on changes to pricing must be made in coordination with other considered changes in resort management, operations, and ski terrain.

Modeling Results & Analysis: Cross Validation

Cross Validation (CV) – Based on the data analysis technique of cross validation (CV), which estimates how well the model is likely to perform on unseen data by testing it across multiple slices of the training data, that further performance in the model is unlikely given variables in excess of eight. This is a statistical observation and cannot be empirically verified.

Further, though such can be considered a valid stand-alone consideration, the following are data deficiencies:

- Assessability of BMR, or how easy is travel to the resort? Serviced by an airport?
- Does the resort accept passes from mountain companies, like Epic, Ikon, or Indy?
- Are there other cultural or seasonal resort events sponsored by Big Mountain?
- What are the operating costs and profit or loss of BMR for the pas five years?

Summary & Conclusions

Summary and Executive Recommendations – We conclude our analysis for considered changes to ticket pricing based on the modeling techniques employed; the expected adult weekend lift-ticket price for the resort is $95.87, compared with its current price of $81.00. This $14.87 pricing gap suggests that the resort may have some flexibility to increase its lift-ticket price. The following changes, or continuing operations, would help to lift ticket pricing:

- We do not recommend making changes to the current number of ski runs.
- Increase the vertical drop by adding a run to a point 150 feet lower, and add two acres of snow making coverage for the additional run.
- Increase the longest run to boast a 3.5 miles length, add snow making coverage.
- Add cultural and/or seasonal resort events, including extreme skiing competition, concerts, and other attractions such as a Gaper Day celebration.